

RACER-GT: A Design-Based Statistical Framework for the Reliable Construction of Long-Horizon Google Trends Time Series

Randomized Acquisition, Global Overlap Calibration,
Generalizability
Decomposition, and Covariance-Adjusted Consensus

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Abstract

Google Trends (GT) is widely used in finance and economics as a high-frequency proxy for investor attention, information demand, and sentiment. Yet the web interface rescales every request to a 0–100 index, a long daily history must be assembled from several short query windows, and repeating an identical query can return different numbers. A researcher who downloads a series once, stitches the windows sequentially, and feeds the result into a financial model therefore inherits four distinct problems at once: segment-scale error, dependence across repeated retrievals, zero inflation and rounding at low search volume, and attenuation bias in any downstream regression.

This paper develops RACER-GT (Randomized Acquisition, Calibration, Error Decomposition, and Reliability for Google Trends). The framework does not claim to recover Google’s absolute search counts. It instead defines two estimands that *are* identified—the expected GT return under a fixed query and a fixed retrieval design, and the latent relative search signal on a common

scale—and then estimates them through eight preregistered stages: (i) a balanced, protocol-locked repeated-retrieval experiment; (ii) global weighted least squares on the overlap graph, which uses every usable overlap simultaneously instead of accumulating error through sequential stitching; (iii) exhaustive retention of exact duplicates together with a residual-based near-duplicate dependence graph; (iv) a three-facet Generalizability Theory decomposition over historical date, collection day, and collection stream; (v) a design-based stratified reference estimator and a covariance-adjusted minimum-variance consensus; (vi) quadratic-programming temporal benchmarking against weekly or monthly aggregates; (vii) separate detection, level, and daily-innovation reliability with a locked acceptance decision tree; and (viii) propagation of the resulting measurement uncertainty into reliability-corrected OLS and SIMEX.

We state the identification conditions explicitly and prove conditional unbiasedness and minimum-variance results under them. Twenty controlled Monte Carlo replications, evaluated at matched information sets, give a mixed result. Cross-pull aggregation cuts mean RMSE against a known latent truth from 7.3519 for a single pull to 3.7421, a 49.1% reduction that improves in all twenty replications, and a further 10.3% below the cross-pull median’s 4.1710. Temporal benchmarking contributes about 0.22 more, holding the estimator fixed ($p \approx 3 \times 10^{-12}$). Covariance-adjusted weighting, however, shows no detectable advantage over the simple cross-pull mean of 3.7253 (-0.0167 , $p = 0.152$). The natural explanation is that dependence is too homogeneous for GLS to exploit, and we test it directly in a four-scenario experiment that puts a subset of pulls behind a shared cache disturbance cutting across the design facets. The explanation fails: GLS wins no heterogeneous scenario significantly and is significantly worse in two. Diagnostics show the constraints are not responsible (the unrestricted solution is identical to the digit) and that the mechanism works as intended (weights correlate -0.562 with shared residual dependence), leaving the error in estimating Σ from nine pulls as what exceeds the efficiency it buys. The theory is untouched — GLS is still BLUE with Σ known — but the practical recommendation is not. None of these figures is a universal optimality guarantee across keywords, regions, or GT versions.

Keywords: Google Trends; investor attention; measurement error; experimental design; generalizability theory; graph calibration; generalized least squares; temporal benchmarking; SIMEX.

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1 Introduction

Search data have become a standard empirical input in finance. Da et al. (2011, 2015) showed that Google search frequency for ticker symbols captures retail investor attention in a way that pre-existing proxies do not; Preis et al. (2013) documented associations between search volume for finance-related terms and subsequent market movements; Choi and Varian (2012) established the broader “predicting the present” agenda. A large applied literature now treats a downloaded GT series as if it were an observed variable measured without error.

That treatment is difficult to defend once the retrieval mechanism is examined. Three features of the GT interface interact badly with long-horizon daily research designs.

Per-request renormalization. Every GT response is rescaled so that the maximum value within the requested window equals 100. The returned numbers are therefore not comparable across requests with different windows. Two requests that differ only in their end date produce two different scales for the same calendar day.

Window-length quantization. Daily granularity is returned only for sufficiently short windows. A multi-year daily history must be requested as a sequence of overlapping chunks and reassembled. The conventional reassembly is sequential: chunk 2 is rescaled onto chunk 1, chunk 3 onto the rescaled chunk 2, and so on. Scale errors then compound multiplicatively along the chain, and the final segment inherits the accumulated error of every preceding join.

Non-determinism. Repeating an identical query on a different day, or from a different browsing environment, can return different values. Eichenauer et al. (2022) document this instability directly and propose averaging across repeated samples. Averaging is a substantial improvement over a single download, but it raises a further question that the applied literature has largely left open: repeated retrievals are *not* independent samples, so what exactly does their average estimate, and how much information do N retrievals actually contain?

1.1 What this paper does

We treat data acquisition itself as a designed, auditable measurement experiment rather than as a preliminary inconvenience. The central move is to refuse the question “what is the true search volume?”, which the GT system does not permit us to answer, and to replace it with two questions that are answerable under stated assumptions:

1. Under a fixed query specification and a fixed, preregistered retrieval design, what is the expected value that GT returns for each historical date?
2. On a common scale, what is the latent relative search signal that the repeated retrievals have in common?

The first estimand is design-based and requires no model of Google’s internal sampling; it is unbiased by construction provided the inclusion rule does not depend on the realized values. The second is model-based and requires the assumptions we make explicit in section 3. Reporting both, and keeping them distinct, is a deliberate methodological choice: the design-based quantity is a transparent benchmark against which the model-based reconstruction can be judged.

1.2 Contributions

1. A preregistered acquisition design. Collection day, collection stream, technical replicate, chunk order, and time slot are balanced and locked before any data are seen, and the entire configuration is committed to a SHA-256 protocol hash (section 4). This makes retrieval-side researcher degrees of freedom auditable rather than invisible.
2. Global overlap-graph calibration. We replace sequential stitching with a single weighted least-squares problem on the graph whose nodes are chunks and whose edges are robustly estimated log-scale differences (section 5). This uses all overlap information at once, removes error accumulation, and yields identification, connectivity, conditioning, and cycle-consistency diagnostics that sequential stitching cannot provide.
3. Dependence accounting rather than deduplication. Exact duplicates are retained in the audit archive and collapsed analytically; near-duplicates are identified from residuals after the common signal is removed, so that highly dependent retrievals are not counted as independent evidence (section 6).

4. A three-facet generalizability decomposition. We estimate variance components for historical date, collection day, and collection stream and their interactions, separately for the level, the detection indicator, and the daily innovation (section 7). The associated D-study answers a question practitioners actually face: whether to buy reliability with more collection days, more streams, or more technical replicates.
5. A covariance-adjusted consensus with an honest effective sample size. Weights solve a minimum-variance problem under the estimated residual covariance, and we report spectral and Kish effective pull counts so that dependence is visible in the reported precision (section 9).
6. Propagation into downstream inference. The measurement uncertainty is carried into reliability-corrected OLS and SIMEX rather than discarded at the boundary of the measurement exercise (section 12).

1.3 Scope and what is deliberately not claimed

RACER-GT estimates a latent *relative* search signal on a common scale. It does not estimate absolute Google search counts, and no result in this paper should be read as identifying them. All unbiasedness, consistency, and minimum-variance statements are conditional on the assumptions listed in section 3, and we state in section 14 exactly which guarantees survive which assumption failures.

2 The measurement problem

2.1 Notation

Let $t \in \mathcal{T} = \{1, \dots, T\}$ index historical calendar dates, and let a *complete pull* i denote one full reconstruction of the historical window under the locked query specification. A complete pull consists of J fixed, overlapping chunks; chunk j covers a contiguous window $W_j \subset \mathcal{T}$, and consecutive chunks overlap in at least $m_{\min}$ days.

Write Y_{ijt} for the raw GT value returned for date t in chunk j of pull i . Each pull is characterized by a design cell: a collection day $d \in \mathcal{D}$, a collection stream $s \in \mathcal{S}$, and a technical replicate $r \in \mathcal{R}$.

Remark 2.1 (Streams are composite environments). *A stream is a fixed combination of device, browser profile, cookie state, account state, network route, and IP address. An estimated stream effect is therefore not an IP effect. Attributing it to the IP address alone would require a crossover factorial design that varies IP while holding the remaining factors fixed. We are explicit about this because the applied literature sometimes treats distinct IP addresses as independent sampling units, which the design does not license.*

2.2 Three failure modes of naive construction

Let X_t denote the latent relative search signal we wish to recover.

(F1) Compounding scale error. Under sequential stitching with estimated log-scale corrections $\hat{\ell}_j$, the calibrated value of chunk J carries the error $\sum_{j=2}^J (\hat{\ell}_j - \ell_j)$. Even if each join is unbiased, the variance of the accumulated error grows with the number of joins, so the oldest or newest part of a long series is systematically less reliable than the middle.

(F2) Dependence masquerading as replication. If N retrievals share a common retrieval-side disturbance, the variance of their mean does not fall as $1/N$. Reporting N as a sample size overstates precision. In the extreme case of exact duplicates, N retrievals contain the information of one.

(F3) Attenuation downstream. If the constructed index enters a regression as a covariate measured with classical additive error of variance σ_u^2 , the OLS slope is attenuated by the reliability ratio $\lambda = \sigma_W^2 / (\sigma_W^2 + \sigma_u^2)$. Measurement work that stops at the construction stage leaves this bias in place.

2.3 What the existing literature establishes

Each failure mode above has been documented separately, and the contribution here is to place them in one measurement chain rather than to discover them. Eichenauer et al. (2022) identify sampling noise and frequency inconsistency in GT and propose repeated draws as the remedy; Cebrián and Domenech (2024) show that the error in repeated pulls scales with a term’s popularity; Medeiros

and Pires (2021) demonstrate that a single realization can flip a forecasting conclusion by chance. On construction, Bleher and Dimpfl (2022) knit multi-annual high-frequency series from overlapping windows, and West (2020) calibrates across queries with an anchor bank that also handles rounding and censoring at low volume. Hölzl et al. (2025) survey how GT is misused in the social sciences, including the practice of reading a constructed index as if it were a count.

Two strands describe the retrieval mechanism itself. Neumann et al. (2023), working with Google Health Trends, document the sampling mechanism and recommend aggregating related phrases with the Boolean OR operator to lift series above the privacy threshold. Most directly related to the present paper, Djorno et al. (2026) treat missing values, sampling variability, noise, and trend as a single preprocessing problem, and report that repeated requests are served from a cache that resets at midnight UTC, so that a fresh sample is drawn only once per day. Their remedy is hierarchical clustering, smoothing splines, and detrending, validated by the accuracy of downstream influenza forecasts.

What that work leaves open is the step from a smoothed curve to an input for inference. None of it fixes the retrieval design in advance as a crossed experiment, decomposes error across collection day and stream, downweights dependent pulls by an estimated residual covariance, or carries the resulting uncertainty into the downstream regression. The distinction from Djorno et al. (2026) is the sharpest available statement of what this paper adds: that work seeks a cleaner curve and is evaluated by whether forecasts improve, whereas the target here is the measurement uncertainty itself, which is estimated, reported, and then propagated. A denoised series with no attached uncertainty is still consumed downstream as if it were measured without error. Once it is not, the downstream problem is the generated-regressor problem of Pagan (1984): a covariate that is itself an estimate makes the conventional standard error wrong even when the point estimate is serviceable. section 12 therefore treats the constructed series as generated rather than observed, in the errors-in-variables tradition of Carroll et al. (2006) and Cook and Stefanski (1994). Those are the gaps this paper addresses, and section 14 states precisely which of them come with proofs and which come with diagnostics only.

Remark 2.2 (Why the website version still needs chunking). *Google documents that Trends normalizes a sample of search activity by time and region and scales it to 0–100, with low volumes reported as zero (Google Trends Help, n.d.). The 2025 API alpha offers a consistent scale across requests, but its historical range is a rolling window of roughly 1,800 days and its values remain relative rather than absolute (Google*

Search Central, 2025). A study spanning 2010–2026 therefore still reconstructs its daily history from multiple shorter windows, which is what makes the calibration problem in section 5 unavoidable rather than optional.

3 Estimands and identification

3.1 Two estimands

Definition 3.1 (Retrieval-mixture expectation). *Fix the query specification Q and a retrieval design that places known probability π_h on design cell $h \in \mathcal{H}$. The retrieval-mixture expectation is*

$$\theta_t^\pi = \sum_{h \in \mathcal{H}} \pi_h \mathbb{E}[Y_t \mid Q, h], \quad (1)$$

where the inner expectation is taken over the GT system’s own randomness conditional on the cell.

Definition 3.2 (Latent common-scale signal). *Let Z_{it} denote the calibrated value of pull i at date t after within-pull scale alignment and baseline normalization. The latent common-scale signal X_t is defined by the measurement model*

$$\mathbf{Z}_t = \mathbf{1}X_t + \mathbf{e}_t, \quad \mathbb{E}[\mathbf{e}_t] = 0, \quad \text{Var}(\mathbf{e}_t) = \Sigma. \quad (2)$$

Definition 3.1 requires no model of the retrieval error and is estimable by stratified averaging. Definition 3.2 requires the assumptions below but supports variance-optimal estimation.

Remark 3.1 (The bridge from θ_t to X_t). *Definition 3.1 and definition 3.2 are not the same target, and two conditions connect them. First, proportionality: $\theta_t = cX_t$ for some $c > 0$ that does not depend on t , so that the expected GT response under a locked protocol tracks the latent relative signal. This rules out a systematic algorithmic bias that varies over time, and repeated measurement cannot test it. Second, negligible calibration error: $Z_{it} = Y_{it} \exp(-\hat{\ell}_{j(i)})$ is a nonlinear transform of an estimate, and $\mathbb{E}[Y \exp(-\hat{\ell})] \neq a^{-1}\mathbb{E}[Y]$ in general because $\exp(\cdot)$ is convex. The first-order correction in section 5 leaves $\mathbb{E}[\exp(-\hat{\ell}_j)] = a_j^{-1}\{1 + O(n^{-1})\}$ with n the number of usable overlap days, so $\mathbb{E}[\mathbf{e}_t] = 0$ in (2) is an $O(n^{-1})$ approximation rather than an exact statement. The unbiasedness and minimum-variance results below should be read under both conditions: the first needs external validation, the second improves as the overlap design is densified.*

Remark 3.2 (Why no additive pull bias is removed by default). *Version 1.0.0 subtracted $b_i = T^{-1} \sum_t (Z_{it} - m_t)$ from each pull, with m_t the daily median. Baseline normalization has already set $T^{-1} \sum_t Z_{it} = 100$ for every pull, so when the baseline spans the full history — the default that QuerySpec back-fills — that centring term reduces to $b_i = 100 - T^{-1} \sum_t m_t$, the same constant for every i . It removes no pull-specific difference; it shifts the entire consensus by the gap between the mean and the median, and it contradicts $\mathbb{E}[e_t] = 0$ in (2). In the controlled simulation it moved mean bias from machine precision to -0.1464 and degraded RMSE in all twenty replications. Version 1.1.0 therefore defaults `center_pull_bias` to off and reports `pull_bias_cross_pull_sd`, which is near zero exactly when the term carries no pull-specific information. Enable it only with baseline rescaling switched off and a genuine additive offset expected.*

3.2 Assumptions

Assumption 3.1 (Protocol invariance). *Every retrieval in a batch uses the identical query specification, historical endpoints, and chunk manifest, and the configuration hash is unchanged across the batch.*

Assumption 3.2 (Value-independent inclusion). *Whether a retrieval is included in the analysis does not depend on the values it returned. Retrievals may be excluded for documented operational reasons recorded before inspection of the values.*

Assumption 3.3 (Positive-overlap multiplicative model). *For a date t present in chunks j and k of the same pull, with both raw values at or above a prespecified threshold $c_{\min}$,*

$$Y_{jt} = a_j X_t \exp(\epsilon_{jt}), \quad a_j > 0, \quad (3)$$

so that with $\ell_j = \log a_j$ and $\nu_{jkt} = \epsilon_{jt} - \epsilon_{kt}$,

$$r_{jkt} = \log Y_{jt} - \log Y_{kt} = \ell_j - \ell_k + \nu_{jkt}. \quad (4)$$

Assumption 3.4 (Connected overlap graph). *The graph $G = (V, E)$ with V the set of chunks and E the set of chunk pairs having at least $m_{\min}$ usable overlapping days is connected.*

Assumption 3.5 (Conditionally centred edge error). *$\mathbb{E}[\nu_{jkt} \mid (j, k) \in E] = 0$ for every edge, and the edge-level estimator is applied to a set of overlap days selected without reference to the realized differences.*

Assumption 3.6 (Common signal with centred retrieval error). *Model (2) holds with Σ finite and positive definite on the retained pulls.*

Remark 3.3 (Where each assumption can fail). *Assumption 3.3 fails near zero, because rounding makes the log ratio explode; this is why only cells at or above $c_{\min}$ contribute to edges, while zeros are never imputed to positive values. One consequence of the aggregation rule must be stated plainly: when a date is covered by several chunks and only some of them return zero, the weighted average is positive, so that date is no longer zero in the final series. Zeros survive only where every covering chunk is zero. That is a corollary of averaging rather than an imputation decision, but it changes what detection reliability is measured on, so `CalibrationResult.diagnostics` reports `n_dates_with_zero_chunk` and `n_dates_zero_masked_by_aggregation` for audit. Assumption 3.4 fails if the chunk design leaves a gap; the implementation raises an error rather than silently returning a partially identified series. Assumption 3.6 fails if a subset of pulls shares a retrieval-side bias that is not shared by the rest; this is precisely what the near-duplicate diagnostics in section 6 are designed to expose.*

4 Stage 0: preregistered acquisition design

The design fixes, before collection begins: the query specification; the historical endpoints; the chunk manifest (window length, step, minimum overlap); the set of collection-day offsets; the set of streams; the number of technical replicates; the time slots; and the acceptance thresholds. Stream order is rotated across collection days and chunk order is rotated across pulls, so that order effects are balanced rather than confounded with the facets of interest.

The entire configuration is serialized canonically and hashed:

$$h = \text{SHA256}\{\text{canonical}(Q, \mathcal{D}, \mathcal{C}, \mathcal{T})\}, \quad (5)$$

where $\mathcal{C}$ collects the calibration settings and $\mathcal{T}$ the decision thresholds. Every raw observation may store h . Any change to a locked setting changes h , so configuration drift between the design and the analysis is detectable rather than assumed away.

Pilot separation. An initial monitoring window may be used to check for gaps, query drift, exact duplicates, and connectivity. If that inspection leads to any change in thresholds, stitching rules, query, code, or decision tree, the monitored

data must be relabelled as pilot and the confirmatory batch restarted. Otherwise the same data are used both to design and to validate the acceptance rule, producing data-dependent acceptance bias. For the same reason, all GT processing rules must be locked before any return, volatility, spread, or VaR result is examined; selecting the construction that gives the most significant financial result converts measurement validation into outcome-guided model selection.

5 Stage 1: global overlap-graph calibration

5.1 Robust edge estimation

For each chunk pair (j, k) with at least $m_{\min}$ usable overlap days, the observed log ratios $\{r_{jkt}\}$ are summarized by a Huber M-estimator of location (Huber, 1964), which solves

$$\sum_t \psi_c \left(\frac{r_{jkt} - \mu}{\hat{\sigma}} \right) = 0, \quad \psi_c(u) = \max\{-c, \min(c, u)\}, \quad (6)$$

with $c = 1.345$ and $\hat{\sigma}$ a robust scale. The estimator returns the location $\hat{r}_{jk}$, a robust scale, and a variance $\hat{v}_{jk}$ that becomes the edge precision. Robustness matters here because a small number of overlap days can be contaminated by revisions or by the rounding boundary, and a sample mean of log ratios is not resistant to those points.

5.2 Graph weighted least squares

Collect the edges into a design matrix B with rows indexed by edges: row e for edge (j, k) has $+1$ in column j , -1 in column k , and zero elsewhere. Because only differences are identified, one chunk is fixed as the reference by setting $\ell_{\text{ref}} = 0$; the implementation selects the highest-degree node, which minimizes the leverage of any single poorly estimated edge. With $\mathbf{r}$ the vector of edge estimates and $\Omega = \text{diag}(\hat{v}_e)$,

$$\hat{\ell} = \arg \min_{\ell} (\mathbf{r} - B\ell)^\top \Omega^{-1} (\mathbf{r} - B\ell) = (B^\top \Omega^{-1} B)^{-1} B^\top \Omega^{-1} \mathbf{r}. \quad (7)$$

Proposition 5.1 (Identification). *Under assumption 3.4, B has rank $|V| - 1$ after the reference normalization, so $\hat{\ell}$ in (7) is uniquely defined.*

Proof. B is the incidence matrix of G . For a graph with $|V|$ nodes, the rank of the incidence matrix equals $|V|$ minus the number of connected components. Connectivity gives one component, hence rank $|V| - 1$. Deleting the reference column removes exactly the one-dimensional null space spanned by $\mathbf{1}$, leaving a full-column-rank matrix and a positive definite normal matrix $B^\top \Omega^{-1} B$. $\square$

Proposition 5.2 (Conditional unbiasedness and efficiency). *Under assumptions 3.3 to 3.5, and treating Ω as known, $\hat{\ell}$ is unbiased for ℓ and is the minimum-variance estimator within the class of linear unbiased estimators of ℓ based on the edge estimates.*

Proof. Write $r = B\ell + \nu$ with $\mathbb{E}[\nu] = 0$ by assumption 3.5 and $\text{Var}(\nu) = \Omega$. This is a linear model with full-column-rank design after normalization, so the Gauss–Markov theorem applies directly to the weighted least-squares estimator (7). $\square$

Remark 5.1 (Why this dominates sequential stitching). *Sequential stitching is the special case in which only the $J - 1$ consecutive edges are used and each is applied in turn. Proposition 5.2 uses every edge, including the non-consecutive overlaps that a long window and a short step generate, and weights each by its estimated precision. It also makes error accumulation impossible: no estimate is defined as a function of a previously calibrated estimate. In addition, the residual $r - B\hat{\ell}$ has a direct interpretation as cycle inconsistency, which is reported as a diagnostic and has no analogue in the sequential procedure.*

5.3 Reconstruction and normalization

Calibrated observations are $\tilde{Y}_{jt} = Y_{jt} \exp(-\hat{\ell}_j)$, optionally with a lognormal bias correction. Overlapping dates are aggregated by inverse-variance weighting. The resulting series is normalized so that its mean over the prespecified baseline window equals 100. This normalization defines the scale of the estimand; it is not an estimate of absolute search volume.

6 Stage 2: duplicate and dependence diagnostics

Two retrievals may be numerically identical, or nearly so. Both cases violate the implicit assumption that repeated pulls contribute independent information, but they require different treatment.

Exact duplicates. Vectors are hashed after rounding to a fixed number of decimals. Exact groups are retained in the audit archive—deleting them would destroy evidence about the retrieval process—and are collapsed analytically before the consensus step, with multiplicity recorded.

Near-duplicates. Raw correlation between two pulls is uninformative, because all pulls share the same strong latent signal and are therefore correlated at close to one by construction. The diagnostic instead removes the common signal first and examines the residual correlation, together with value-anchored measures: mean absolute deviation on the 0–100 scale, exact cell agreement, and the Jaccard index of the positive-value sets. Pairs that exceed the prespecified thresholds form the edges of a dependence graph.

Remark 6.1 (Components are diagnostic, not equivalence classes). *Connected components of the dependence graph are reported as connectivity sets. Membership in a component does not imply that every pair inside it exceeds the threshold, because connectivity is transitive while the pairwise rule is not. The acceptance rule uses the share of the batch occupied by the largest component, which is the quantity that actually matters for whether the batch is dominated by one dependence cluster.*

7 Stage 3: generalizability decomposition

Classical reliability coefficients assign all non-signal variation to a single undifferentiated error term. Generalizability Theory (Cronbach et al., 1972; Brennan, 2001) instead decomposes it across the facets of the measurement design, which is what a researcher needs in order to decide where to spend the next unit of collection effort.

For a fully crossed design over historical date t , collection day d , stream s , and replicate r ,

$$Y_{tdsr} = \mu + T_t + D_d + S_s + (TD)_{td} + (TS)_{ts} + (DS)_{ds} + (TDS)_{tds} + E_{tdsr}. \quad (8)$$

Variance components are estimated by the method of moments from the balanced ANOVA sums of squares. The implementation validates full crossing and equal cell counts before estimating, and raises an error otherwise, rather than silently producing components from an unbalanced design.

Remark 7.1 (Confounding with a single replicate). *When $n_r = 1$, E cannot be separated from (TDS) ; the implementation retains the combined term and reports it as such. This is stated rather than hidden because it changes the interpretation of the residual component.*

The decomposition is run on three transformations, which answer three different questions:

- level, Y itself: do relative magnitudes reproduce?
- detection, $\mathbb{I}\{Y > 0\}$: does the zero/non-zero pattern reproduce? This is the binding constraint for low-volume terms.
- innovation, $\Delta \log(1 + Y)$: do daily *changes* reproduce? This is the relevant question when the index enters a model in differences, and it is systematically harder to satisfy than the level criterion.

The generalizability coefficient for relative decisions and the dependability coefficient for absolute decisions follow in the usual way, and the D-study evaluates them on a grid of hypothetical designs, so the marginal value of an additional collection day can be compared with that of an additional stream.

8 Stage 4: the design-based reference estimator

Let h index the prespecified collection-day $\times$ stream cells with target weights π_h summing to one, and let n_h be the number of completed pulls observed in cell h . The stratified estimator is

$$\hat{\theta}_t^\pi = \sum_h \pi_h \frac{1}{n_h} \sum_{i \in h} Z_{it}. \quad (9)$$

Proposition 8.1 (Design unbiasedness). *Under assumptions 3.1 and 3.2, and provided $n_h \geq 1$ for every cell with $\pi_h > 0$, $\hat{\theta}_t^\pi$ is unbiased for the retrieval-mixture expectation θ_t^π of definition 3.1.*

Proof. Conditional on cell h , the retrievals in that cell are draws from the GT system under a fixed specification, so $\mathbb{E}[n_h^{-1} \sum_{i \in h} Z_{it}] = \mathbb{E}[Y_t \mid Q, h]$ by assumption 3.2, which guarantees that inclusion carries no information about the values. Taking the π -weighted sum over cells and applying linearity gives (1). $\square$

Remark 8.1 (Dependence affects the variance, not the bias). *Proposition 8.1 does not require the retrievals to be independent. Dependence across pulls inflates $\text{Var}(\hat{\theta}_t^\pi)$*

but leaves the expectation unchanged. This is exactly why the estimator is retained as a transparent reference: its validity rests on the design, not on a covariance model that could itself be misspecified.

9 Stage 5: covariance-adjusted consensus

9.1 Minimum-variance weights

Under assumption 3.6, consider linear estimators $\hat{X}_t = \mathbf{w}^\top \mathbf{Z}_t$ with $\mathbf{w}^\top \mathbf{1} = 1$, which is the condition for unbiasedness.

Proposition 9.1 (BLUE weights). *Under assumption 3.6 with Σ known and positive definite,*

$$\mathbf{w}^* = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}} \quad (10)$$

solves $\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$ subject to $\mathbf{w}^\top \mathbf{1} = 1$, and the resulting $\hat{X}_t = \mathbf{w}^{\top} \mathbf{Z}_t$ has variance $(\mathbf{1}^\top \Sigma^{-1} \mathbf{1})^{-1}$, the minimum over all linear unbiased estimators.*

Proof. The Lagrangian is $L = \mathbf{w}^\top \Sigma \mathbf{w} - 2\lambda(\mathbf{w}^\top \mathbf{1} - 1)$ with first-order condition $\Sigma \mathbf{w} = \lambda \mathbf{1}$, hence $\mathbf{w} = \lambda \Sigma^{-1} \mathbf{1}$. Imposing $\mathbf{w}^\top \mathbf{1} = 1$ gives $\lambda = (\mathbf{1}^\top \Sigma^{-1} \mathbf{1})^{-1}$ and (10). The objective is strictly convex because $\Sigma \succ 0$, so the stationary point is the unique minimum. Substituting yields the stated variance. $\square$

9.2 Stabilization in finite samples

Proposition 9.1 assumes Σ known. In practice the number of pulls is of the same order as, or smaller than, the number of dates used to estimate the covariance, so the sample covariance is ill-conditioned and its inverse is unstable. The implementation therefore:

1. collapses exact duplicates analytically, so that a repeated vector cannot inflate the apparent rank;
2. removes a pull-level offset, so that a constant retrieval bias is not mistaken for signal disagreement;
3. estimates Σ by Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) toward a well-conditioned target;
4. imposes $w_i \geq 0$ and $w_i \leq \bar{w}$ while retaining $\sum_i w_i = 1$.

Remark 9.1 (What the constraints cost). *The nonnegativity and cap constraints are stabilization devices, not improvements on proposition 9.1. The strict BLUE property belongs to the unconstrained solution. The constrained solution is a minimum-variance convex combination: it gives up the theoretical optimum in exchange for refusing the large offsetting positive and negative weights that an ill-conditioned $\hat{\Sigma}^{-1}$ would otherwise produce. We regard this as the right trade for a measurement instrument that must behave predictably on data the analyst has not yet seen.*

9.3 Effective number of pulls

Because dependence is the central concern, the raw count of pulls is not reported as a sample size. We report instead the Kish effective count (Kish, 1965) $1/\sum_i w_i^2$ and a spectral effective count derived from the eigenvalues of the residual correlation matrix. Both fall below the nominal count when retrievals are dependent, and the acceptance rule in section 11 is stated in terms of the spectral count.

The spectral count carries a known downward bias that should be read alongside it. Residuals are formed by subtracting the daily median *of the pulls themselves*, which induces negative correlation among them, so the residual correlation matrix is not the identity even when the underlying pulls are independent. On independently generated series the estimate averages 8.57 at $m = 9$ and 2.59 at $m = 3$, shortfalls of 4.8% and 13.7%. The direction is conservative — it understates information and therefore errs toward rejecting a batch — but it means the effective strictness of `min_spectral_effective_pulls` exceeds its nominal value, most visibly in small designs.

10 Stage 6: temporal benchmarking

Long GT histories at daily frequency are not directly comparable to the weekly or monthly series that GT returns for the same query over the full window. Let z be the preliminary daily consensus, A the aggregation matrix mapping days to benchmark periods, and b the lower-frequency benchmark. The soft benchmark solves

$$\min_x (x - z)^\top Q (x - z) + (Ax - b)^\top W (Ax - b), \quad (11)$$

where $Q = (\lambda_I + \lambda_R)I + \lambda_D D_1^\top D_1$ combines fidelity to the daily consensus with a *first-difference* smoothness penalty on the correction path, and W is the bench-

mark precision. The ridge λ_R is a separate configuration field whose only role is to keep Q positive definite when $\lambda_I = 0$, so the two identity terms are one coefficient mathematically. The default $\lambda_D/\lambda_I = 10$ is a strong smoothing prior: it pushes corrections toward low-frequency adjustments and can systematically damp daily shape near event days, which is where an attention index is most informative. Calibrate and lock that ratio in a pilot, and report sensitivity over $\lambda_D/\lambda_I \in \{1, 10, 100\}$. This is the quadratic-programming form of the movement-preservation idea familiar from Denton (1971) and Chow and Lin (1971); treating b as noisy rather than exact is appropriate here because the lower-frequency GT series is itself a sampled quantity.

Proposition 10.1 (Existence and uniqueness). *If $Q \succ 0$ and $W \succeq 0$, the objective in (11) is strictly convex and the solution is the unique $x^* = (Q + A^\top W A)^{-1}(Qz + A^\top W b)$.*

Proof. The Hessian is $2(Q + A^\top W A)$, which is positive definite because $Q \succ 0$ and $A^\top W A \succeq 0$. Setting the gradient to zero gives the stated normal equations, whose coefficient matrix is invertible. $\square$

11 Stage 7: reliability and the acceptance decision tree

Three reliability questions are kept separate because they can have different answers for the same series: whether the zero/non-zero pattern reproduces (detection, summarized by Fleiss' κ , Fleiss, 1971), whether levels reproduce, and whether daily innovations reproduce.

The acceptance decision is a locked tree evaluated after estimation. Each rule compares an observed diagnostic against a prespecified threshold and returns pass, fail, or indeterminate. The batch status is

- FAIL if any mandatory rule fails;
- REVIEW if no mandatory rule fails but at least one is indeterminate;
- PASS otherwise.

The rules cover protocol integrity, overlap-graph connectivity, the number of numerically distinct pulls, the spectral effective pull count, the share of zero observations, level and innovation generalizability, detection agreement, concentration of the near-duplicate dependence graph, consensus convergence as the last pull is added, and consistency with the lower-frequency benchmark.

Remark 11.1 (Indeterminate is not a soft fail). *The detection rule is not identified when every observation falls in one detection category, because κ is undefined without variation. The implementation reports the rule, marks it indeterminate, and—crucially—makes it non-mandatory in exactly that case, so that a series with no zeros is not penalized for a coefficient that cannot exist. Conflating “cannot be computed” with “failed” would systematically reject high-volume terms.*

Remark 11.2 (What PASS does and does not mean). *PASS authorizes construction under the locked protocol. It does not establish that the keyword has construct validity for the economic concept of interest, it does not identify absolute search counts, and it licenses no causal interpretation.*

12 Stage 8: propagating measurement error downstream

Let $W_t = X_t + u_t$ be the constructed index, used as a regressor for an outcome Y_t with controls. Under classical additive measurement error, the moment-corrected estimator is

$$\hat{\beta} = \frac{\mathbf{W}^\top \mathbf{M} \mathbf{Y}}{\mathbf{W}^\top \mathbf{M} \mathbf{W} - \text{tr}(\mathbf{M} \Omega_u)}, \quad (12)$$

where $\mathbf{M}$ is the residual maker for the controls and Ω_u the measurement error covariance supplied by the construction stage. Standard errors use a HAC estimator (Newey and West, 1987). The implementation raises an error when the corrected denominator is non-positive, rather than returning an explosive or sign-flipped coefficient; a non-positive denominator means the estimated measurement error exceeds the observed regressor variation, which is a statement about the data, not a numerical detail to be smoothed over.

As an alternative that does not require the classical structure to hold exactly, SIMEX (Cook and Stefanski, 1994; Carroll et al., 2006) adds further known noise on a grid of λ values, records the resulting estimates, fits an extrapolation function, and evaluates it at $\lambda = -1$.

13 Controlled Monte Carlo evidence

The simulation generates a latent signal with realistic dynamics, applies collection-day and stream effects, adds correlated retrieval error, splits the history into independently normalized chunks, imposes rounding and zeros, and optionally injects

exact duplicates. Because the latent truth is known by construction, estimators can be compared directly.

13.1 Matching the information sets

`PipelineResult.final_series` returns the benchmarked series whenever a benchmark is available. Comparing that series against cross-pull baselines that never received the benchmark does not compare four estimators; it compares three estimators against one estimator plus an extra block of low-frequency information, and in simulation that block is generated from the latent truth itself. Every estimator is therefore evaluated twice: on the calibrated pulls alone, and after applying the *same* weekly and monthly benchmark. The single-pull row reports mean performance over all pulls rather than one fixed pull, because a fixed column measures which pull happened to sort first, not what a researcher who downloads once should expect.

Table 1: Twenty controlled replications (seeds 1200–1219). MCSE is the Monte Carlo standard error of the mean RMSE. Lower RMSE is better.

Estimator	Benchmark	RMSE	MCSE	Corr.	Innov. corr.	Peak recall
Single pull	no	7.3519	0.0573	0.9688	0.8878	0.8912
Cross-pull median	no	4.1710	0.0819	0.9898	0.9556	0.9447
Simple mean	no	3.7253	0.0877	0.9919	0.9747	0.9526
RACER-GT	no	3.7421	0.0894	0.9918	0.9743	0.9526
Single pull	yes	6.9881	0.0491	0.9717	0.8882	0.8947
Cross-pull median	yes	3.9584	0.0695	0.9908	0.9558	0.9474
Simple mean	yes	3.5058	0.0754	0.9928	0.9748	0.9553
RACER-GT	yes	3.5226	0.0770	0.9927	0.9745	0.9526

Table 2: Paired comparisons on RMSE. A positive difference favours estimator A; $n = 20$ paired replications.

Estimator A	Estimator B	Difference	t	p	A wins
RACER-GT	Single pull	+3.6098	48.31	2.4×10^{-21}	20/20
RACER-GT	Cross-pull median	+0.4289	14.86	6.5×10^{-12}	20/20
RACER-GT	Simple mean	−0.0167	−1.49	0.152	7/20
RACER-GT (bm.)	Simple mean (bm.)	−0.0168	−1.65	0.116	7/20
RACER-GT (bm.)	RACER-GT (no bm.)	+0.2194	15.49	3.1×10^{-12}	20/20
Simple mean (bm.)	Simple mean (no bm.)	+0.2195	15.45	3.3×10^{-12}	20/20

Three conclusions follow, two of which support the framework and one of which does not. First, cross-pull aggregation buys a great deal: at a matched information

set RACER-GT cuts RMSE by $1 - 3.7421/7.3519 = 49.1\%$ against a single pull, improving in all twenty replications, and by a further 10.3% against the cross-pull median. Second, covariance-adjusted weighting shows no detectable advantage over the simple mean, before benchmarking (-0.0167 , $p = 0.152$) or after it (-0.0168 , $p = 0.116$). Third, temporal benchmarking contributes about 0.22 on its own, and contributes it almost identically to RACER-GT and to the simple mean (0.2194 versus 0.2195): the gain belongs to the benchmark, not to the daily estimator it is paired with.

Remark 13.1 (A testable prediction). *The second conclusion is a negative result, and its most natural explanation is that dependence in this DGP is homogeneous: every pull shares its collection-day noise with fixed weight 0.5, so the residual covariance is close to equicorrelated, the minimum-variance weights of an equicorrelated matrix are equal weights, and GLS has nothing to exploit while still paying for estimating Σ .*

If that explanation holds, GLS should win once dependence strength differs across pulls — when some pulls are served from one cache and others are not. That is a testable prediction, and section 13.2 tests it directly.

13.2 A direct test of dependence heterogeneity

`examples/run_dependence_experiment.py` places a subset of pulls behind one shared cache disturbance. Membership cuts across collection days and streams, so the extra dependence *cannot* be absorbed by the design facets and is visible only in the residual covariance — which is what the GLS weights exist for. At `cache_cluster_fraction = 0` the simulated data are byte-identical to those behind table 1, so the two sections are directly comparable.

Table 3: Four dependence scenarios, 20 replications each. Advantage is simple mean minus GLS in RMSE, so positive favours GLS.

Scenario	GLS	Simple mean	Median	Advantage	MCSE	p	GLS wins
Homogeneous (baseline)	3.7421	3.7253	4.1710	-0.0167	0.0112	0.152	7/20
1/3 of pulls, moderate	3.7658	3.6801	4.1133	-0.0857	0.0155	< 0.0001	2/20
1/3 of pulls, strong	4.1684	4.1980	4.6486	$+0.0296$	0.0439	0.508	10/20
1/2 of pulls, strong	5.1633	4.8802	5.5593	-0.2831	0.0591	0.0001	2/20

The prediction fails. GLS wins no scenario significantly. The closest case (one third of pulls, strong dependence) is $+0.0296$ with $p = 0.51$ and exactly 10 wins out of 20, which is a tie; in two scenarios GLS is *significantly worse*. Adding heterogeneity does not reverse the ranking.

Two diagnostics rule out the obvious misreadings of that result.

First, the constraints are not the cause. Relaxing them under the strong one-third scenario (10 replications), the constrained default, the uncapped variant, and the fully unrestricted solution all attain a mean RMSE of 4.1752 — identical to the digit, because the unrestricted solution already lies inside the feasible set. An empirical covariance gives 4.1785 and a diagonal one 4.2345, against 4.2415 for the simple mean over the same seeds. The unrestricted estimator, which is BLUE in theory, does not win either.

Second, the mechanism works. The correlation between a pull’s mean residual correlation and the weight it receives is -0.562 , negative in all ten replications: GLS does find the pulls that share dependence and does move weight away from them. Weight dispersion in table 3 rises monotonically with heterogeneity ($0.0220 \rightarrow 0.0409$) and the spectral effective pull count falls monotonically ($4.93 \rightarrow 2.95$); both correctly register the added dependence.

What remains is estimation error. Σ must be estimated from 9 pulls over 366 historical dates, and the perturbation that its error induces in the weights exceeds the efficiency that exploiting the true dependence structure can buy. This matches a long-standing finding in the portfolio literature, where sample-covariance minimum-variance portfolios fail to beat equal weighting in small samples; Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) was proposed to mitigate exactly this, and is already the default here.

Remark 13.2 (What the negative result changes, and what it does not). It does not change the theory. With Σ known, the GLS weights remain minimum-variance among linear unbiased estimators (proposition 9.1); that derivation depends on no simulation. It changes the practical recommendation: at the design sizes this paper recommends (21–42 pulls, a single year), no detectable gain in point accuracy should be expected from the covariance adjustment, and a simple mean is an equally defensible primary choice.

It also does not change the diagnostic value of the adjustment. The spectral effective pull count fell from 4.93 to 2.95, correctly telling the researcher that nine pulls are now worth about three. A simple mean cannot supply that judgment, nor the near-duplicate dependence structure, the per-day conditional standard errors, or measurement uncertainty in a form the downstream EIV correction can consume. RMSE cannot answer whether a batch is usable at all, and that is what this framework is mainly for.

Remark 13.3 (Reading the bias column). Mean bias for the single pull and for the

simple mean sits at the 10^{-15} level. That is an identity, not an estimator property: every complete pull is normalized to baseline mean 100 and the truth is defined on the same baseline, so their average cannot be biased. RACER-GT now lands at machine precision as well ($+3.2 \times 10^{-17}$). The cross-pull median's -0.1464 is real: a daily median sits systematically below the mean under the right-skewed event spikes in this DGP.

14 Scope of the statistical guarantees

We state precisely what is and is not established.

1. Under assumptions 3.1 and 3.2, the stratified design-cell average is design-unbiased for the prespecified retrieval-mixture expectation (proposition 8.1). This holds *without* any independence assumption across pulls.
2. Under assumptions 3.3 to 3.5, the relative log-scale parameters are identified and the graph WLS estimator is conditionally unbiased and efficient among linear estimators built on the edge estimates (propositions 5.1 and 5.2).
3. Under assumption 3.6 with Σ known or consistently estimated, the unconstrained GLS weights are minimum-variance among linear unbiased estimators (proposition 9.1). The constrained default is a stabilized convex combination and does not inherit this optimality.
4. Given a correct lower-frequency benchmark and an identified constraint set, the benchmark estimator is the unique minimizer of an explicit quadratic objective (proposition 10.1).
5. Under classical additive measurement error with a consistent estimate of Ω_u , (12) corrects attenuation bias.

None of these results establishes unbiasedness with respect to Google's absolute search counts, and none is claimed to hold unconditionally.

14.1 Known limitations

- The variance components use balanced method-of-moments estimation. REML for unbalanced designs is a planned extension.
- Consensus confidence intervals are conditional approximations. A full-pipeline moving-block bootstrap—resampling historical blocks and re-estimating edges,

scales, duplicates, covariance, weights, and the benchmark before re-estimating the financial model—is the appropriate procedure for confirmatory inference.

- Benchmark noise, scale-calibration uncertainty, and tuning-parameter uncertainty are not yet fully propagated into the reported intervals.
- Zero values may mix genuine zero search, privacy thresholds, sampling, and rounding. Version 1.2.0 retains zeros and isolates detection reliability, but does not implement an interval-censored latent-count likelihood. For high zero-share terms, moving to weekly frequency or to a broader Topic is usually more defensible than manufacturing daily values.
- Anchor-bank calibration across different queries is documented as an extension but is not implemented in version 1.2.0.
- The package is deliberately ingestion-first and embeds no unofficial scraper, because endpoints, rate limits, authentication, and terms of service change. Collectors need only emit the documented long-format schema.

15 Conclusion

Google Trends is a measurement instrument with a documented tendency to return different numbers for the same question. Treating its output as an error-free observation is not a conservative simplification; it silently imports scale error, dependence, and attenuation into whatever model consumes it.

RACER-GT takes the opposite stance. It fixes the retrieval design before any data are seen, calibrates all chunks simultaneously on the overlap graph, measures rather than assumes the independence of repeated retrievals, decomposes the remaining variation across the facets of the design, and carries the resulting uncertainty forward into downstream estimation. What it delivers is not certainty about Google’s internal counts, which is unavailable, but a reproducible, auditable, and explicitly conditional estimate of a well-defined target, together with a locked rule for deciding whether a given batch of data is fit to use at all.

The controlled evidence is deliberately mixed. Under this data-generating process, cross-pull aggregation and temporal benchmarking both deliver large and robust reductions in reconstruction error, while covariance-adjusted weighting does not beat a simple mean. We report the comparison that does not favour us alongside the ones that do, because that is the same discipline the framework asks

of its users: a measurement procedure that can only produce flattering numbers cannot tell a researcher when to stop trusting it.

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